From Persistent Disagreement to Agreement: Learning with Revisions of Perceptual Sets

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ABSTRACT

We often view sensation and perception as one process and mistakenly take our internal representation of the physical world as the world itself. Cognitive biases could occur in the perception process without our noticing. This paper zooms in on the learning process and highlights a specific subprocess—the perception process. Agents take the same sensory information as inputs, direct them through different *perceptual sets* and thus form different *perceptual information*, which is then used to update posterior beliefs. Biased perceptual sets thus lead to persistent disagreement among agents, while *set revision* corrects any perceptual bias and allows agreement to be reached. Depending on the intensity of *perceptual shocks*, agents' sensitivity to perceptual shocks and their initial *entrenchment level*s, there could be overreaction or underreaction in the price to sensory information in the markets. Perceptual revisions can induce rebalancing, while a volume response requires an equilibrium portfolio specification.

Keywords: perceptual bias, cognitive frictions, persistent beliefs

JEL codes: D82, D83, G11, G14, G41.

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1 Introduction

The world overflows with information that is visual, auditory, olfactory, gustatory, and etc. In order to make sense of all this *external* information, our body has developed a series of processes for sensing and perceiving the outside world. The first step is *sensation*, "the process of detecting external events with sense organs and turning those stimuli into neutral signals" (Krause et al., 2017). The sensory information is then relayed to the brain where the second step, *perception*, occurs. Perception is "the organization, identification, and interpretation of sensory information" (Schacter et al., 2014) in order to form an *internal* representation of the world. As an example, when investors are flipping through business news in the *Wall Street Journal*, the sensory receptors in their eyeballs are registering different patterns of light reflecting off the pages, converting them to neutral signals which are then integrated and processed by the brain into meaningful perception of words. However, humans are not machines; what is conveyed physically on the paper is likely not what readers understand.

According to Bruner and Postman (1949), when encountering new information, we are open to different cues. However, as we process more information, some cues become familiar and thus selective. We search for more cues that confirm the categorization of the target. We also actively ignore and even distort cues that violate our initial perceptions. Our perception becomes more selective and we finally re-create a consistent internal presentation of the new information. In other words, we interpret patterns of stimuli through a *perceptual set*—"a readiness to perceive a stimulus in a particular way" (Weiten, 2001).

An important factor that heavily influences our perception is our expectation (Bruner and Minturn, 1955), which is characterized by the prior belief in this chapter. Perceptual sets are also affected by motivational forces—people have a tendency to see what they want to see (Balcetis and Dunning, 2006), and the selection of sensory input—people tend to perceive what they focus their attention on (Chun and Wolfe, 2005) and overlook unexpected items and events that do not receive their attention (Simons and Chabris, 1999). Another factor that affects perception sets

is *confirmation bias*, that is people seek information that supports their prior beliefs and ignore information that contradicts them (Nickerson, 1998). Confirmation bias is also widely studied in the finance literature.

Motivational factors, selective perception and confirmation bias all vary from person to person; it is akin to "individual fixed effect" and thus summarized in this chapter by the blanket term *personal traits*. These personal traits make perception sets very persistent—it is hard for people to change how they view the world. Nevertheless, it is not completely impossible to achieve that, especially after disastrous events. Those events are so intense that people's attention is drawn to them even if it was placed elsewhere and that contradictions to prior beliefs is too significant to overlook. In other words, disastrous events post *perceptual shocks* that suddenly and greatly change perceptual sets.

Individuals process the same *sensory information* through different perceptual sets and thus form different *perceptual information*, that is different internal representation of the same external inputs, and then combine them with the prior belief to form posterior beliefs[1]. The processes of perception and belief updating are shown in Figure 1.

To model dynamic belief updating and jumps of perceptual sets, I introduce the tools that are frequently used in continuous-time finance. Models of utility in stochastic continuous-time settings typically assume that beliefs are represented by a probability measure. For example, Chen et al. (2024) models two competitive agents, one with the objective belief in the sense that her probability measure is consistent with the physical measure, and the other with a probability measure that is different from but equivalent to the physical measure. In this chapter, I model perceptual sets as probability measures, while beliefs about the unobservable process are obtained through Kalman filter. Individuals are not aware of the existence of perceptual sets; how the external information is converted to internal information is a blackbox to them. Starting from a common prior, individuals observe a random process $\{P_t\}$ and try to make inferences about a

[1]In psychology, belief is defined as the mental acceptance of something to be true (Weiten, 2001), which is consistent with its economic interpretation as a probability distribution over a set of possible states.

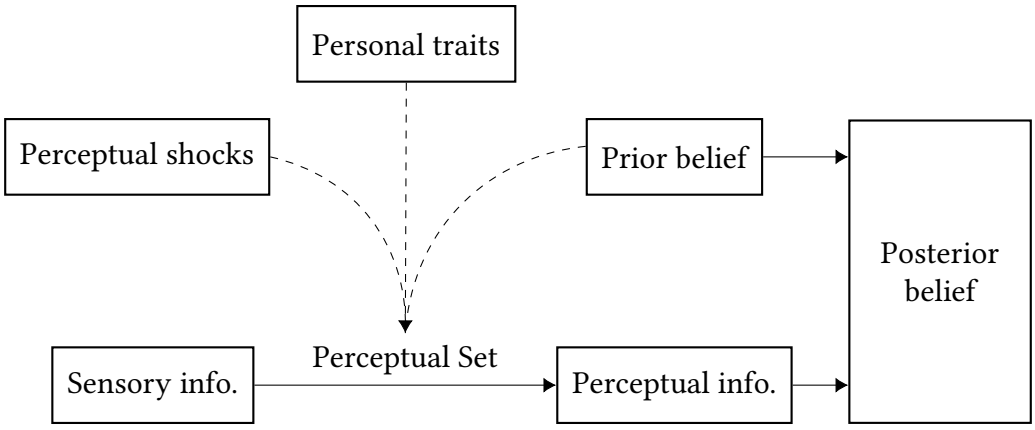


Figure 1. Sensory information, perception and belief

I abstract from sensation and directly take sensory information as raw input. It goes through the perception process, during which it is filtered by a subjective perceptual set and converted to *perceptual information*. It is then combined with the prior belief to form the posterior belief. The perceptual set is determined by personal traits (motivational factors, selective perception and confirmation bias), the prior belief and perceptual shocks. Perceptual shocks could make the perceptual set subject to sudden and big changes.

function of P_t and an unobservable random process $\{\theta_t\}$. The inference is de facto done under individuals' subjective probability measures. In the spirit of entrenchment for beliefs[2], I define a notion of *set entrenchment*. A perceptual set with low entrenchment is more subject to revision. Agents are endowed with an initial set entrenchment level. Arrivals of independent perceptual shocks and shocks from P_t will affect entrenchment. The perceptual set is revised when the entrenchment level hits the set revision threshold. Even with common prior and information, individuals have persistently different beliefs. In somes cases, we also observe side-switching.

Another interesting question to think about is to disentangle the effects of new information on the posterior belief versus changes in the posterior belief resulted from perceptual set revision. The first one is the standard Bayesian updating, while the second one has the flavor of Collin-Dufresne et al. (2016), which implemented parameter learning for model selection. This chapter takes a different angle, in that independent perceptual shocks could lead to perceptual set revision, resulting a different posterior despite of the same old information. New information could be absent in this setting.

[2]In belief contraction, the beliefs with the lowest entrenchment should be the ones that are most readily given up (Hansson, 2017).

This general theoretical framework of subjective perceptual sets could be applied to different areas in economics and finance. We can introduce it to the financial markets to explain excess volatility and trading volume. It also explains why investors initially underreact to certain news but suddenly the market perspective is changed, and why tail events happen more often than common wisdom suggests. On the micro-level, if investors are aware of the existence of the perceptual set but do not know how it is biased, what can they do to improve their utility? And can a social planner devise any policy to increase investors' utility and stablize the market? The framework can also be used to explain belief polarization in political economy and bubbles in household expectations.

Literature There is a growing literature that tries to explain persistent disagreement among agents. The starting point of many studies is a well-known result by Savage (1954). He argued that repeated observation of signals will lead a Bayesian agent to assign probability 1 to a true event almost surely. What we observe, however, is frequently at odds with this standard theoretical framework; persistent disagreement among agents still exists despite of abundant information.

To explain this observation, some research has been done under the rational inattention framework. For example, in Nimark and Sundaresan (2019), agents can choose channels, which are noisy, costly, and have different precisions in different states of the world, to observe the reality. They allocate more precision to those states a priori more likely, which perpetuates differences in beliefs. Kominers et al. (2018) study Bayesian agents for whom computing posterior beliefs is costly; such agents face a tradeoffs between economizing on attention costs and having more accurate beliefs. Even small processing costs can lead to significant departures from the standard costless processing model.

However, humans are not machines; we process information in different ways than a machine does. Psychologists and cognitive scientists have shown that individuals display biases in their perception of the world, upon which another strand of research in economics is based. In that literature, it is offen assumed that agents do not know certain state of the world *and* the quality

of information regarding the state. For example, Cheng and Hsiaw (2022) drew on the *curse of knowledge*, that is individuals who have seen information often fail to ignore it when the context requires using only prior beliefs, to model *pre-screeners* who "double-dip" the data. Pre-screeners firstly form a first-stage belief about credibility based on newly-observed signals, and then use it instead of their prior to weight all signals. This proposed mistake predicts when individuals over- or underreact to new information, depending on the order in which they received previous signals, and explains why information that should objectively mitigate disagreement may amplify it instead. In a similar vein, Koçak (2018) showed that if a signal is informative on multiple dimensions, and if individuals can hold only a limited number of items in working memory, that is after the agent updates the first belief, she has to discard the prior of the now updated parameter, and use its posterior and the new information to update the second parameter, then the order with which beliefs are updated across different dimensions leads to different posteriors. This provides an underlying mechanism of why the curse of knowledge comes into effect.

In contrast, in this chapter individuals do not doubt the credibility of experts but process sensory information in an *unnoticeably* biased way. Individuals do not question the quality of information source, nor do they make active information acquisition. They are given a full set of information, but each individual process information under a different perceptual set, resulting persistent disagreement. The perceptual set is not just a result from experiences or prior information, but could also be biological, e.g. sensitivity to different types of sensory information. Thus individuals can be free of double-dipping the data but still updating information in a biased way.

In Chauvin (2023), a dissonance-reducing learner wishes to agree with his sources of information, and therefore distorts his subjective perception of the arguments before adding them to the prior. Distortion is achieved through scaling the arguments by real-valued factors. Fryer et al. (2019) modeled "double updating" when agents are faced with ambiguous information. The agent first interpret each signal based on their current belief and then form a posterior on the sequence of interpreted signals. In this chapter, however, belief polarization happens even with unambiguous information. This chapter also extends the idea of subjective perception of param-

eters/information by introducing a general perceptual set.

2 Learning without perceptual bias

Although perceptual sets are generally shaped by personal traits, previous perceptual shocks and prior beliefs, they are very persistent. Once they are formed, the influence they receive from belief updating through combining new information is minimal. Revisions of perceptual sets occur mainly following huge perceptual shocks, which are unforeseeable rare events. In the absence of perceptual set revisions, the learning process can be modelled by optimal linear filtering. The agents are not aware of the perception process and bias, and the perceptual shocks are exogenous, unforeseeable and rare. Once perceptual shocks hit and the perceptual set is revised, the agent repeats the learning process and apply the optimal linear filtering under her new perceptual set. In this section, I introduce the general framework of optimal linear filtering, which is joined piecewise once the perceptual set revision is induced exogenously.

Let (V_t, P_t), $t \in [0, T]$ be a two-dimensional random process, with V_t being the unobservable state of the world that is to be learned, and P_t is observable. More specifically, P_t and V_t are related by

$$\mathrm{d}P_t = A_t V_t \,\mathrm{d}t + B_t(P_t)\,\mathrm{d}Z_t = \left[A_t V_t - \eta_t^i\right]\mathrm{d}t + B_t(P_t)\,\mathrm{d}Z_t^i, \tag{1}$$

where A_t is deterministic at time t, and $B_t(P_t)$ is $\mathcal{F}_t^P$-measurable. We are interested in the optimal[3] estimator for V_t. Assume that V_t can be represented by

$$\mathrm{d}V_t = a_t V_t \,\mathrm{d}t + b_t \,\mathrm{d}Z_t^f\,. \tag{2}$$

where a_t and b_t satisfy $\mathbb{P}\left(\int_0^T |a_s V_s|\,\mathrm{d}s < \infty\right) = 1$ and $\int_0^T a_s^2 \mathbb{E}^i\left[V_s^2\right]\mathrm{d}s < \infty$; b_t is a random process with $\mathbb{P}\left(\int_0^T |b_s|\,\mathrm{d}s < \infty\right) = 1$, and Z_t^f is a standard Wiener process over $(\Omega, \mathcal{F}, \mathbb{P})$, independent from Z_t. Let $\mathcal{F}_t^P = \{P_t; 0 \le s \le t\}$ denote the information filtration generated by the history of signal realizations, which is publicly observable. Agents also have a common prior

[3] It is optimal in the minimum mean-square error sense.

$V_0 \mid \mathcal{F}_0^P \sim N(m_0, \gamma_0)$. The conditional mean of V_t from agent i's perspective, which is summarized by the following proposition.

PROPOSITION 1. *Let* $m_t^i \equiv \mathbb{E}^i \left[V_t \mid \mathcal{F}_t^P \right]$. *Then*

$$\mathrm{d}m_t^i = a_t m_t^i \, \mathrm{d}t + \frac{A_t \gamma_t^i}{B_t(P_t)} \, \mathrm{d}Z_t^{Pi}, \tag{3}$$

where $\mathrm{d}Z_t^{Pi} = \frac{1}{B_t(P_t)} \left[\mathrm{d}P_t - (A_t m_t^i - \eta_t^i) \, \mathrm{d}t \right]$.

See the appendix for proofs and the framework for a more general learning objective. By applying $h_t = V_t^2$ to Theorem 1 and Ito's Lemma to Proposition 1, we can get the tracking error (conditional variance) of V_t from agent i's perspective, which is summarized by the following proposition.

PROPOSITION 2. *Let* $\gamma_t^i \equiv \mathbb{E}^i \left[(V_t - m_t^i)^2 \mid \mathcal{F}_t^P \right]$. *Then the conditional variance follows the Riccati equation*

$$\mathrm{d}\gamma_t^i = \left(2 a_t \gamma_t^i + b_t^2 - \frac{A_t^2 (\gamma_t^i)^2}{B_t(P_t)^2} \right) \mathrm{d}t \, . \tag{4}$$

This equation, together with (3), characterizes the linear-Gaussian filter.

3 Learning under perceptual bias

3.1 Perceptual sets

Consider the complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Assumes that there is a continuum of measure one ex-ante identical behavioral agents who live over a finite time interval $[0, T]$. Agent i has a *subjective* perceptual set, which is implemented by probability measure $\mathbb{P}^i$, different from but equivalent to the *objective* perceptual set, that is the physical measure $\mathbb{P}$. The two measures are related by

$$\mathrm{d}Z_t^i = \mathrm{d}Z_t + \frac{\eta_t^i}{B_t(P_t)} \, \mathrm{d}t, \tag{5}$$

where η_0^i is the initial perceptual bias, measured in units of signal drift. It only changes when set revision is triggered; otherwise it remains constant. As a benchmark, I refer to a comparable Bayesian agent who always has an objective perceptual set, that is she always process sensory information perfectly and thus not subject to set revision.

3.2 Revision of perceptual sets

3.2 Simplistic modelling of a rare disaster

At a fixed time t, agent i receive an exogenous perceptual shock, which changes their perceptual bias η_0^i by a multiplier q, $q \geq 0, q \neq 1$. When $q = 0$, agent i is no longer perceptually biased. When $0 < q < 1$, she becomes less perceptually biased. When $q > 1$, she becomes more perceptually biased.

This simplistic modelling of perceptual set revision is suitable for characterizing a rare disaster event. To see a clear-cut effect of a disaster, the smaller effects of non-perceptual shock factors that can affect perceptual set revision are normalized to zero. The large magnitude of the unpredicted disaster rationalizes the setting of a perceptual set revision.

3.2 Set entrenchment

More general scenarios allow perceptual shocks, personal traits and prior beliefs to jointly affect perceptual sets. To achieve that, I introduce set entrenchment level into the framework. At time $t = 0$, agent i has an initial set entrenchment level E_0^i. The observed observation of P_t not only take part in belief updating as the source of new information, but also affect the set entrenchment level. Normally, the latter effect is small and could be positive or negative. The change in set entrenchment level is predominantly affected by independent perceptual shocks, which always reduce the set entrenchment level, if occur. Mathematically, let E_t^i be the set entrenchment level for agent i at time t, follows: Let N^i be a Poisson process with intensity λ^i, with iid marks $\beta_n \in (0, 1]$, independent of the Brownian motions. Let $\rho_0^i = 0$ and, recursively, let ρ_{k+1}^i be the first shock arrival $T_n^i > \rho_k^i$ at which the process below is nonpositive. On each interval

$[\rho^i_k, \rho^i_{k+1})$, entrenchment is

$$E^i_t = E^i_0 + \alpha^i \int_{\rho^i_k}^{t} B_s(P_s)\,\mathrm{d}Z_s - \zeta^i \sum_{n:\rho^i_k < T^i_n \le t} \beta_n.$$

where ζ^i is the sensitivity to perceptual shocks and T^i_n is the nth shock arrival. At each revision time ρ^i_{k+1}, the agent revises her perceptual set towards the objective view according to $\eta^i_{\rho^i_{k+1}} = (1 - \beta_{N^i_{\rho^i_{k+1}}})\eta^i_{\rho^i_{k+1}-}$. The next interval starts from E^i_0, so the jump that triggered the reset is not applied again. Perceptual shocks therefore create actual downward jumps in entrenchment; regardless of the sign of the initial bias, a revision shifts the bias towards zero.

3.3 Results

In this section, I first present the special case where the state of the world does not change, and then a more general case with a stochastic world state.

3.3 Special case: time-invariant state of the world

Consider the special case where the unknown state of the world V is time-invariant, and the observable P_t is simplified to $\mathrm{d}P_t = V\,\mathrm{d}t + \sigma\,\mathrm{d}Z_t$. Individuals observe $\{P_s : 0 \le s \le t\}$ to make inference for V. Without the revision of perceptual shocks, the conditional expectation and variance of the state variable can be solved from equations 3-4, and the results are shown as follows.

Proposition 3. *In the special case where $A_t = 1$, $B_t(P_t) = \sigma$, and the state of the world V is time-invariant, consider the Bayesian agent and the behavioral agent in the absence of perceptual set revision.*

1) The time-t conditional mean of V from the Bayesian agent's view is

$$m_t = \frac{m_0\sigma^2 + \gamma_0(P_t - P_0)}{\gamma_0 t + \sigma^2}. \tag{6}$$

2) The behavioral agent's view differs by

$$m_t^i = m_t + \eta^i \frac{\gamma_0 t}{\gamma_0 t + \sigma^2}. \tag{7}$$

3) All agents agree on the conditional variance of the underlying state:

$$\gamma_t^i = \gamma_t = \frac{\gamma_0 \sigma^2}{\gamma_0 t + \sigma^2}. \tag{8}$$

4) The Bayesian agent eventually learns about the true state of the world V, while behavioral agent i eventually thinks the state of the world to be $V + \eta^i$.

In this special case, all agents, regardless of their perceptual sets, share the same conditional variance of the state variable, which is decreasing in t. The conditional expectations, however, vary with the perceptual bias. At a fixed time, perceptual shocks are imposed. The conditional moments can be calculated piecewise. After the perceptual shocks, agents still agree with the conditional variance of the state variable, the conditional expectation still varies by the remaining perceive bias.

If the intensity of or agents' sensitivity to perceptual shocks is large enough, revisions of perceptual sets occur for all agents. Further assume that the perceptual bias is completely corrected once perceptual set revision occurs, then the conditional expectation of the state variable viewed by all agents converge to the true value of the state variable. As an illustration, Figure 2 plots the empirical distribution of the conditional expectation of the state variable and the empirical distribution of the entrenchment level at time T. In this special case, the true value of V is set to be 1, and the default entrenchment level is normalized to 1. Tests do not reject the normal distribution hypothesis for either panel.

Figure 3 plots the random paths of the conditional expectation of the underlying state. Absent of perceptual shocks, the conditional expectations change smoothly from updating with new information. When exogenous perceptual shocks are introduced at a fixed time ($t = 10$), perceptual sets are revised and thus conditional expectations experience jumps. The magnitude of

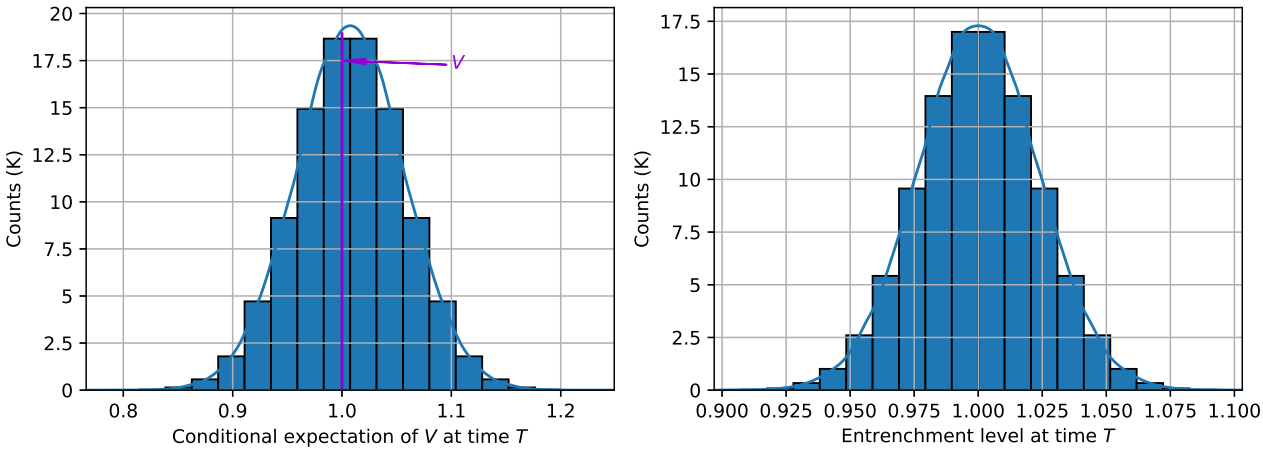


Figure 2. Distributions of conditional expectations of V and entrenchment levels

The left panel plots the empirical distribution of $m_T^i = \mathbb{E}^i[V \mid \mathcal{F}_T^\xi]$ with $V = 1$ and $\eta^i \sim N(0,1)$. The true value $V = 1$ is also noted on the plot. The right panel plots the empirical distribution of E_T^i with the default entrenchment level, which is the level to which entrenchment adjusts after a perceptual set revision, normalized to 1. Tests do not reject the normal distribution hypothesis for either panel.

the jumps reflects the effects of perceptual set revision, isolated from any incoming new information. Assume the sensitivity to perceptual shocks and the intensity of perceptual shocks are large enough, then the perceptual bias is immediately eliminated. The conditional expectations of the underlying state by all agents converge to its true value.

3.3 General case

Generally, the unknown state of the world V_t is a random process, as specified by (2). To retain the exact linear-Gaussian filter, suppose the coefficients are known and deterministic, and individuals observe $\{P_s : 0 \leq s \leq t\}$ to make inference for V_t. A compatible specification of (V_t, P_t), $t \in [0, T]$, is:

$$\mathrm{d}V_t = rV_t\,\mathrm{d}t + \sigma_v\,\mathrm{d}Z_t^f, \tag{9}$$

$$\mathrm{d}P_t = A_tV_t\,\mathrm{d}t + B_t\,\mathrm{d}Z_t, \qquad B_t > 0, \tag{10}$$

where r, σ_v, A_t, and B_t are known. Thus this subsection takes $B_t(P_t) = B_t$, the deterministic-coefficient specialization of the preceding filtering framework.

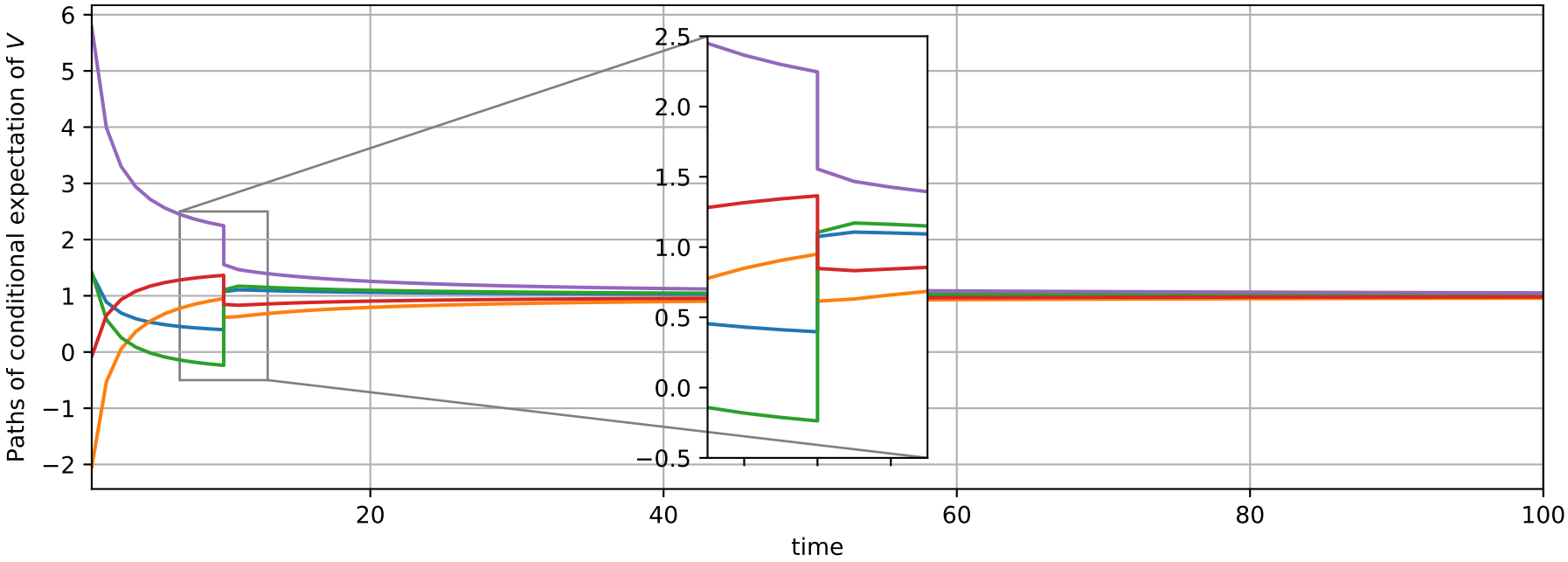


Figure 3. Paths of conditional expectations of θ with perceptual set revisions

Five random paths of m_t^i, the conditional expectation of the underlying state, are plotted. When exogenous perceptual shocks are introduced at time $t = 10$, perceptual sets are revised and thus conditional expectations experience jumps. The magnitude of the jumps reflects the effects of perceptual set revision, isolated from any incoming new information.

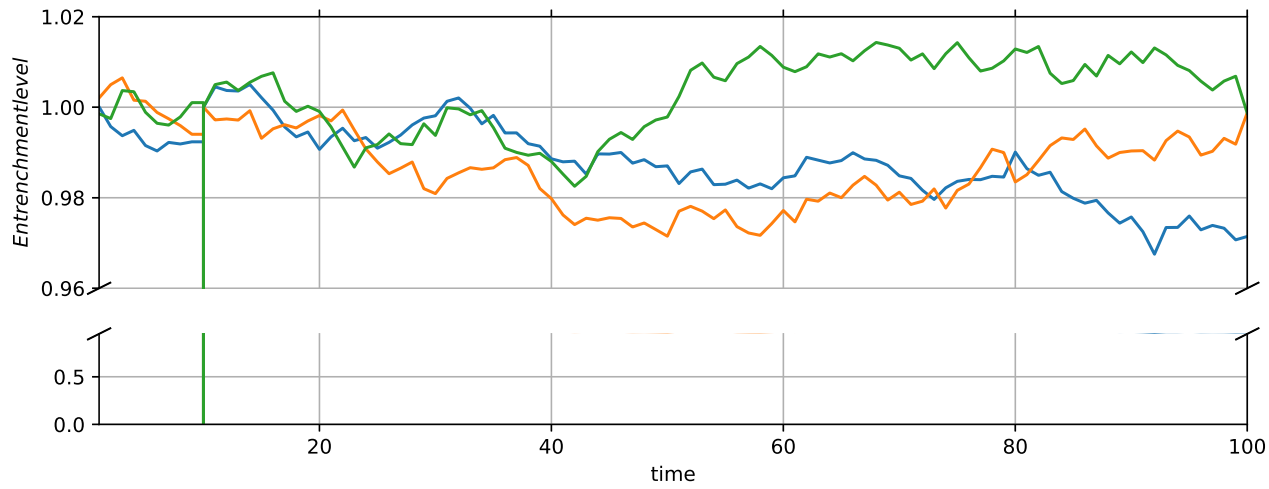


Figure 4. Paths of set entrenchment levels

Three random paths of the entrenchment level E_t^i are presented in this figure. Perceptual shocks are introduced at $t = 10$. Regardless of the entrenchment level at that time, it drops to 0 immediately and thus revision of perceptual sets occurs. The entrenchment level is then reset to the default value 1. When there are no perceptual shocks, the changes in the entrenchment level are very minor, insufficient to accumulate for other revisions of perceptual sets.

Let $\mathcal{F}_t^P = \{P_s; 0 \leq s \leq t\}$ denote the information filtration generated by the history of equilibrium price which are publicly observable by every investor. At time t, from investor i's perspective, the asset value V_t has conditional mean $m_t^i = \tilde{\mathbb{E}}^i \left[V_t \mid \mathcal{F}_t^P \right]$ and variance $\gamma_t^i = \tilde{\mathbb{E}}^i \left[(V_t - m_t^i)^2 \mid \mathcal{F}_t^P \right]$. The innovation process is $\mathrm{d}Z_t^{Pi} = \frac{1}{B_t} \left[\mathrm{d}P_t - (A_t m_t^i - \eta_t^i)\, \mathrm{d}t \right]$.

Figure 4 shows three random paths of the entrenchment level. Perceptual shocks are introduced at $t = 10$. In the illustrated calibration, the shock is large enough to bring entrenchment to its revision threshold, after which the next interval starts at the default value 1. Without a threshold crossing, a shock need not trigger a revision.

Figure 5 plots two sets of perceptual bias and entrenchment level paths. The initial (default) entrenchment level is normalized to 1. The perceptual bias is only revised when the entrenchment level drops to 0, after which the entrenchment level will reset to its default value 1. The entrenchment level changes slightly over time through belief updating but normally these small changes do not result in a revision of the perceptual set. In both panels, only at $t = 10$ when an exogenous perceptual shock is introduced does the perceptual set undergoes revision.

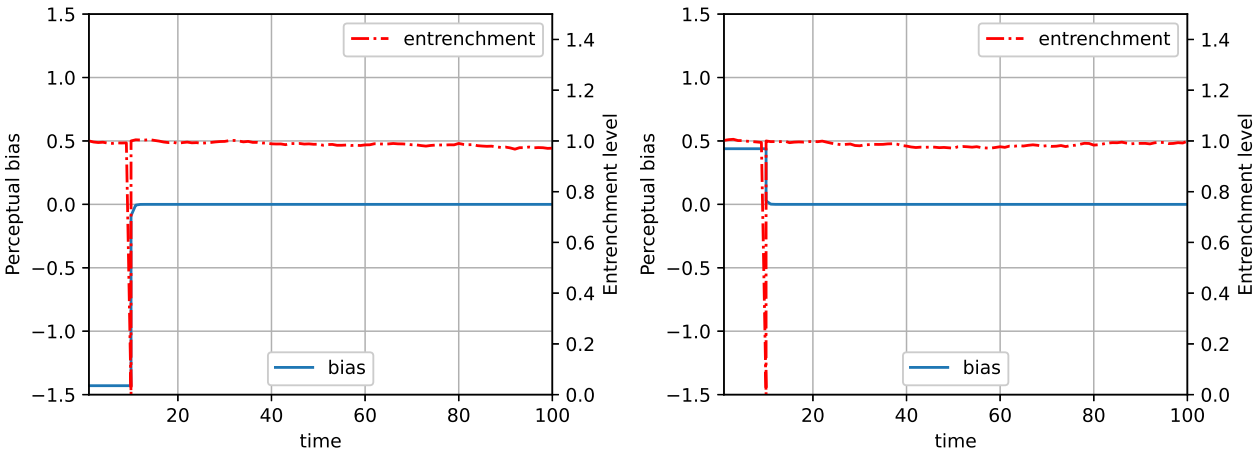


Figure 5. Examples of perceptual bias and entrenchment paths

The two panels plots two sets of perceptual bias and entrenchment level paths. The initial (default) entrenchment level is normalized to 1. The perceptual bias is only revised when the entrenchment level drops to 0, after which the entrenchment level will reset to its default value 1. The entrenchment level changes slightly over time through belief updating but normally these small changes do not result in a revision of the perceptual set. In both panels, only at $t = 10$ when an exogenous perceptual shock is introduced does the perceptual set undergoes revision.

3.3 Experience and set entrenchment

It is reasonable to think that experience affects entrenchment reduction. On the one hand, more experienced investors could initially hold their set more affirmatively. In other words, they will have higher initial set entrenchment level E_0^i. This is in a similar vein to Jovanovic and Nyarko (1996), which argued that an agent may be so skilled at a technology that he will refuse to switch to a better but unfamiliar one. On the other hand, more experienced investors could be better skilled at dealing with perceptual shocks, and thus they reduce set entrenchment at a higher rate, that is they have a larger ζ^i.

3.3 Overwhelming information and sensitivity to set shock

When the private signal about dividend growth rate is more precise, that is when σ_ξ is small, investors have more information to take in. The effects of having more information are two-fold. The good side is that more information helps investors make more informed portfolio choices. However, there is also a bad side. When new information becomes too overwhelming, investors become less sensitive to perceptual shocks, thus less likely to revise their sets.

4 Connecting with the financial markets

The modelling of learning with entrenchment sets can be connected with the financial markets. In Chen et al. (2024), investors "agree to disagree"; belief is modeled as a random process, while information and learning are not explicitly modeled. Chen and Epstein (2002) discussed risks and asset returns when an investor is using the wrong probability measure. With a continuous-time intertemporal version of *multiple-priors utility*, their model delivers restrictions on excess returns that admit interpretations reflecting a premium for risk and a separate premium for ambiguity. This chapter takes a different perspective in that I model information and learning explicitly, and distinguish beliefs and perceptual sets, where the latter are entrenched and hard to change.

4.1 Market setup

The perceptual bias model is readily re-interpretable as investors having perceptual biases towards the underlying value of the risky asset in the financial markets. Consider a continuous-time endowment economy over a finite time interval $[0, T]$. Investors have the same perceptual set structure as in the previous sections. In this section, I introduce assets into the model and specify the unknown state of the world and signal structures.

Assume that there is a riskless bond in the market with a constant risk-free rate $r > 0$. The process P_t remains the public signal governed by (10); it is not identified with a tradable price in this reduced-form portfolio illustration. There is 1 share of aggregate stock with underlying value V_t and tradable price S_t, whose return is exogenously specified by

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}\underset{t}{Z}\,.$$

Here Z_t^f is the fundamental shock to the underlying value and Z_t is the public shock entering both the signal and the reduced-form return.

Investor i has initial wealth W_0 and can invest in both the risk-free asset and the stock. Her

wealth dynamics is

$$\mathrm{d}W_t^i = \left(rW_t^i - C_t^i\right)\mathrm{d}t + \theta_t^i S_t(\mu - r)\,\mathrm{d}t + \theta_t^i S_t \sigma\,\mathrm{d}Z_t,$$

where μ and σ are exogenously given instantaneous return parameters for this reduced-form portfolio problem, distinct from the signal coefficients (A_t, B_t). The first term is the return earned on W_t^i if investor i invests solely in the risk-free market, and the second term is the total excess return earned from the θ_t shares of the stock. Equivalently,

$$\mathrm{d}W_t^i = \left[\left(r + \pi_t^i(\mu - r)\right)W_t^i - C_t^i\right]\mathrm{d}t + \pi_t^i \sigma W_t^i\,\mathrm{d}Z_t,$$

where $\pi_t^i = \frac{\theta_t^i S_t}{W_t^i}$ is the proportion of wealth invested in shares of stock.

Investor i has a time-additive separable utility, given by $\tilde{\mathbb{E}}_0\left[\int_0^T e^{-\rho s}u(C_s^i)\,\mathrm{d}s + e^{-\rho T}U(W_T^i)\right]$, where C is the consumption, ρ is investor's subjective discount rate, $u(C)$ is utility from the consumption flow, and $U(W_T^i)$ is the utility over terminal wealth. At time t, investor i's objective is to seek the optimal investment and consumption strategy to maximize the expected utility given by

$$F\left(t, W_t\right) = \max_{\pi, C} \tilde{\mathbb{E}}^i\left[\int_t^T e^{-\rho(s-t)}u(C_s^i)\,\mathrm{d}s + e^{-\rho(T-t)}U(W_T^i) \mid \mathcal{F}_t^{\xi}, P_t, S_t\right].$$

Notice that when investor i makes decisions, she take expectations under the measure $\mathbb{P}^i$, not the risk-neutral measure $\mathbb{P}$, since she is unaware of the existence of perceptual bias.

For this reduced-form return specification, the value function satisfies

$$0 = \max_{\pi, C}\left\{u(C) - \rho F + \frac{\partial F}{\partial t} + \left[\left(r + \pi(\mu - r)\right)W - C\right]\frac{\partial F}{\partial W} + \frac{1}{2}\pi^2\sigma^2 W^2 \frac{\partial^2 F}{\partial W^2}\right\}. \tag{11}$$

4.2 Results

In the reduced-form illustrations, perceptual revisions can generate abrupt changes in posterior beliefs. A complete equilibrium linking these changes to overreaction or underreaction requires a price-clearing condition derived from the learning model. On the one hand, if the initial entrenchment level is high or the sensitivity to perceptual shocks is low, the entrenchment level would not drop to 0, which leads to an underreaction to shocks. On the other hand, overreaction could happen when perceptual sets are revised if investors have accumulated enough old sensory information. When investors revise their perceptual sets, previous sensory information would go through the new perceptual set to form revised perceptual information, which is then used to correct their posterior beliefs and previous underreaction to news. This is different from belief updating on new information. It is the effects of perceptual set revision with old information.

The plotted price paths are illustrative monotone transformations of beliefs, not equilibrium prices obtained from market clearing.

In a model with trading, volume can be defined by aggregating the absolute value of $\mathrm{d}\theta_t^i$. A perceptual revision changes investors' posterior beliefs and can induce rebalancing, but a volume increase requires equilibrium portfolio demands and is not implied by the filter alone. Figure 6 illustrates paths of a price proxy and the underlying value. Perceptual shocks are introduced at $t = 10$, indicated by a vertical dashed line. If their intensity or agents' sensitivity is sufficiently large, the shocks induce a sudden jump in conditional expectations. The upper and lower panels illustrate belief movements conventionally labelled overreaction and underreaction; they are not equilibrium price results.

5 Conclusion

This chapter zooms in on the learning process. The sensation process transforms external inputs into sensory information, which is subsequently directed through the perceptual process

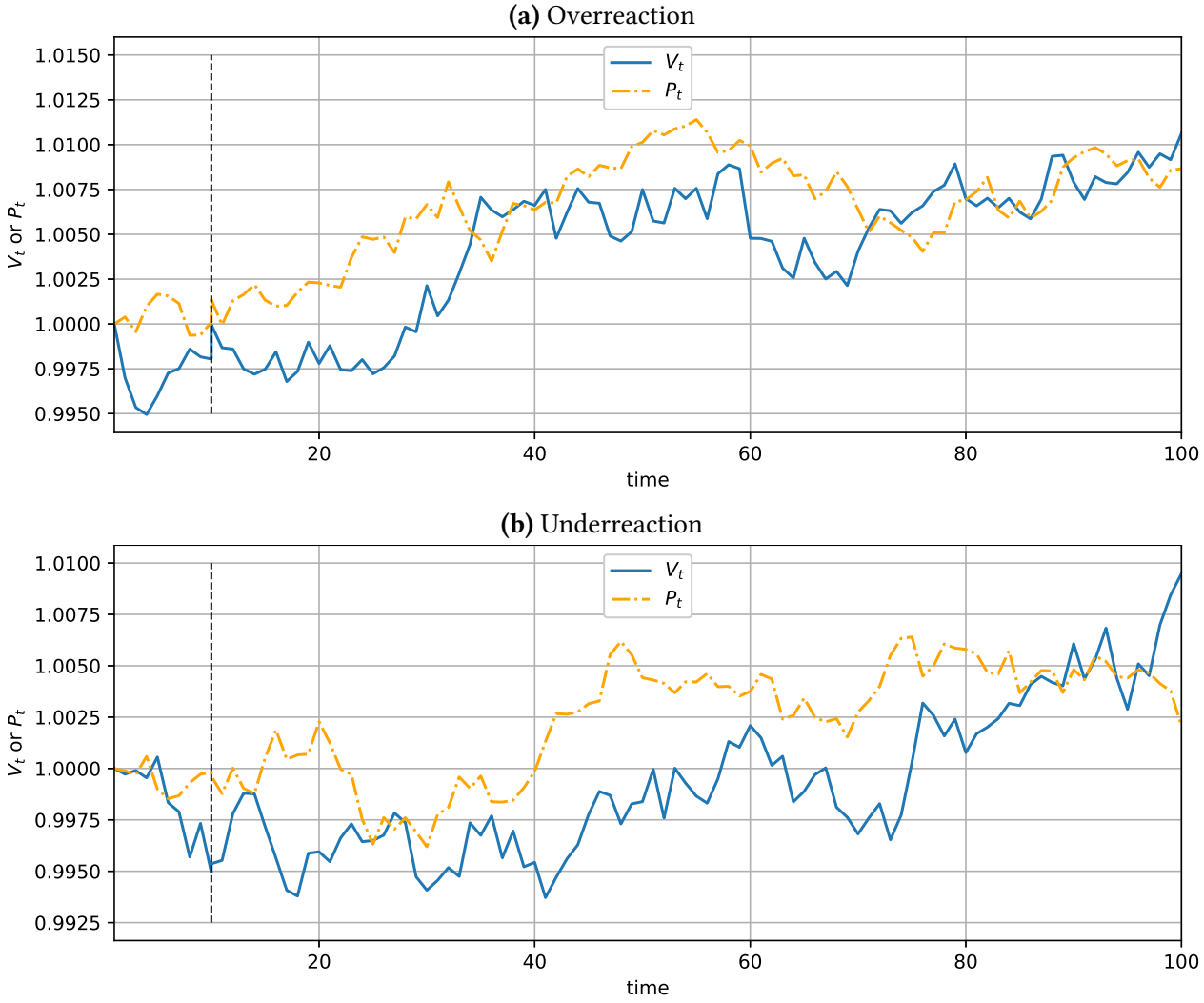


Figure 6. Evolutions of the price and value of the risky asset

This figure plots the paths of the price and value of the risky asset. The upper panel plots an example of overreaction to perceptual shocks. Perceptual shocks are introduced at $t = 10$, indicated by a vertical dashed line. If the intensity of or the sensitivity to perceptual shocks are sufficiently large, perceptual shocks induce a sudden jump in investors' conditional expectations of the risky asset. As a result, an overreaction to new information is reflected in the price. The lower panel plots an example of underreaction to perceptual shocks.

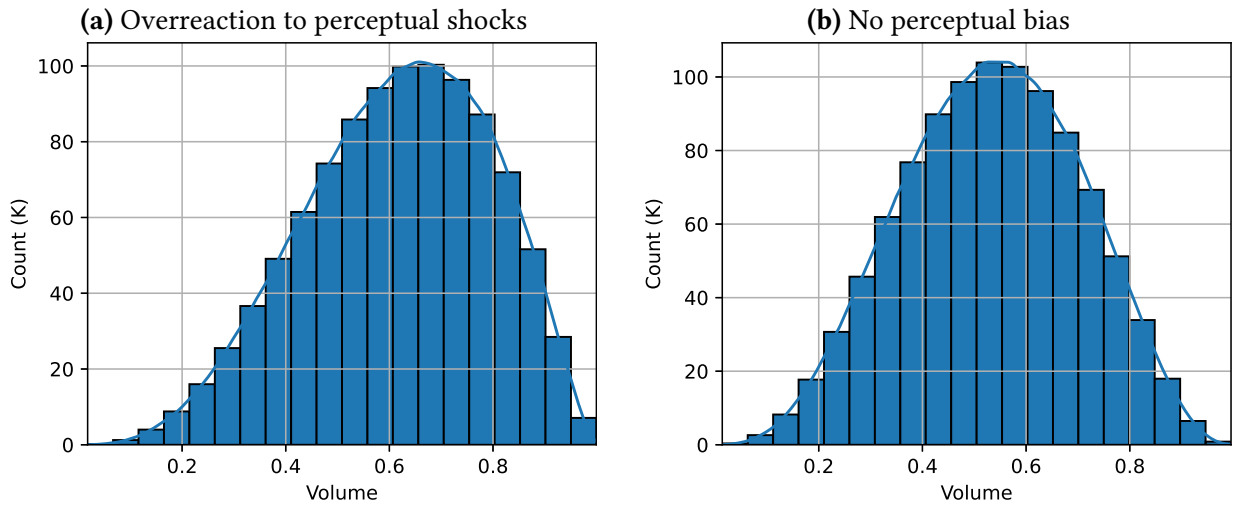


Figure 7. Distributions of trading volumes

This figure plots the histograms of the total trading volumes for the risky asset. In the left panel, perceptual shocks hit and result in an overreaction via perceptual set revisions. As investors adjust their perceptual sets, posterior beliefs are corrected under the new perceptual sets using all old and new information. The average total trading volume of the risky asset thus increases. The right panel plots the trading volumes in the case where there is no perceptual bias.

to generate perceptual information. Perceptual information is then used to update prior beliefs to posterior beliefs. This chapter highlights the perception process and models it by perceptual sets. Agents have biased perceptual sets. Its entrenched nature is modelled by set entrenchment; once formed, the entrenchment level is hard to change. When perceptual shocks occur, however, the enacted perceptual sets would suddenly change and the entrenchment level would be reset to the default value, that is agents become entrenched with their new perceptual sets.

With perceptual bias, even after sufficient information is assimilated, there is still persistent disagreement among agents. Revision of perceptual sets induced by perceptual shocks shifts disagreement to agreement. The magnitude and speed of shift are affected by agents' experience and sensitivity to perceptual shocks. The entrenched nature of perceptul sets and sudden set revision in the face of huge shocks allow room for abrupt movements in beliefs. Whether these movements imply overreaction, underreaction, or higher trading volume depends on the equilibrium price and portfolio-demand specification.

The perceptual set mechanism can be applied to many other scenarios, for example, belief

polarization in political economy. Traditionally, political polarization affects finance through two channels. Through the partisan utility channel, partisans obtain utility through from engaging with their fellow party members or from owning specific financial securities. My perceptual set mechanism is close to the partisan beliefs channel. Partisans may hold different beliefs regarding the state of the economy, expected risk-adjusted returns in the cross section of stocks, or the ability of in-group versus out-group partisans. The notion of a *partisan perceptual screen* was introduced in a seminal study by Campbell et al. (1960). Heterogeneity in beliefs could arise from differences in information environments (e.g., via partisan media), differences in attention, or differences in individuals' models of the world.

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Appendix

A General framework of optimal linear filtering

Generally, the unobservable state of the world V_t and the observable signal P_t are related by

$$\mathrm{d}P_t = A_t V_t \,\mathrm{d}t + B_t(P_t)\,\mathrm{d}Z_t = \left[A_t V_t - \eta_t^i\right]\mathrm{d}t + B_t(P_t)\,\mathrm{d}Z_t^i, \tag{12}$$

where A_t is deterministic at time t, and $B_t(P_t)$ is $\mathcal{F}_t^P$-measurable. We are interested in the optimal (minimum mean-square error) estimator for an $\mathcal{F}_t^P$-measurable function $h_t(V_t, P_t)$. Assume that h_t can be represented by

$$\mathrm{d}h_t = H_t\,\mathrm{d}t + b_t\,\mathrm{d}Z_t^f, \tag{13}$$

where H_t is a random process with $\mathbb{P}\left(\int_0^T |H_s|\,\mathrm{d}s < \infty\right) = 1$ and $\int_0^T \mathbb{E}^i\left[H_t^2\right]\mathrm{d}t < \infty$, b_t is a random process with $\mathbb{P}\left(\int_0^T |b_s|\,\mathrm{d}s < \infty\right) = 1$, and Z_t^f is a standard Wiener process over $(\Omega, \mathcal{F}, \mathbb{P})$, independent from Z_t. Let $\mathcal{F}_t^P = \{P_t; 0 \le s \le t\}$ denote the information filtration generated by the history of signal realizations, which is publicly observable. Agents also have a common prior $V_0 \mid \mathcal{F}_0^P \sim N(m_0, \gamma_0)$. Let $\pi_t^i(h) \equiv \mathbb{E}^i\left[h_t \mid \mathcal{F}_t^P\right]$. Theorem 1 shows how $\pi_t^i(h)$ is updated.

THEOREM 1. *Let the partially observable random process (V, P) permit the representation given by (12) and (13). If $\sup_{0\le t\le T}\mathbb{E}^i[h_t^2] < \infty$ and $\int_0^T \mathbb{E}^i\left[V_t^2\right]\mathrm{d}t < \infty$, then $\forall t \in [0, T]$, it holds almost surely with respect to probability measure $\mathbb{P}^i$ that*

$$\mathrm{d}\pi_t(h) = \pi_t(H)\,\mathrm{d}t + \frac{A_t}{B_t(P_t)}\left[\pi_t(hV) - \pi_t(h)\pi_t(V)\right]\mathrm{d}Z_t^{Pi}, \tag{14}$$

where the innovation process $\mathrm{d}Z_t^{Pi} = \frac{1}{B_t(P_t)}\left[\mathrm{d}P_t - (A_t\pi_t(V) - \eta_t^i)\,\mathrm{d}t\right]$.

Theorem 1 can be proved by invoking the Liptser-Shiryaev optimal filtering equation[4] and change of measure. However, it does not explicitly specify the dynamics that V_t follows. In some

[4]Liptser and Shiryaev (2001) proved a more general version of the optimal filtering equation.

special cases, though, we can write out its dynamics. If $h_t = V_t$ and $H_t = a_t V_t$, then

$$\mathrm{d}V_t = a_t V_t \,\mathrm{d}t + b_t \,\mathrm{d}Z_t^f \,. \tag{15}$$

Then equation (12) and (15) are the state-space representation that is widely studied in economics and finance literature. Applying Theorem 1, one can get the conditional mean of V_t from agent i's perspective, which is summarized by the following proposition.

Proposition 4. *Let* $m_t^i \equiv \mathbb{E}^i \left[V_t \mid \mathcal{F}_t^P \right]$. *Then*

$$\mathrm{d}m_t^i = a_t m_t^i \,\mathrm{d}t + \frac{A_t \gamma_t^i}{B_t(P_t)} \,\mathrm{d}Z_t^{Pi}, \tag{16}$$

where $\mathrm{d}Z_t^{Pi} = \frac{1}{B_t(P_t)} \left[\mathrm{d}P_t - (A_t m_t^i - \eta_t^i) \,\mathrm{d}t \right]$.

For the linear-Gaussian system, the tracking error (conditional variance) of V_t is summarized by the following Riccati equation.

Proposition 5. *Let* $\gamma_t^i \equiv \mathbb{E}^i \left[(V_t - m_t^i)^2 \mid \mathcal{F}_t^P \right]$. *Then*

$$\mathrm{d}\gamma_t^i = \left(2 a_t \gamma_t^i + b_t^2 - \frac{A_t^2 (\gamma_t^i)^2}{B_t(P_t)^2} \right) \mathrm{d}t \,. \tag{17}$$

B Solutions to the main model

Proof of Proposition 5. For a scalar linear-Gaussian state-space model, the Kalman–Bucy covariance equation is the Riccati equation in (17). It follows by applying Itô's formula to $\mathbb{E}^i[(V_t - m_t^i)^2 \mid \mathcal{F}_t^P]$ and using the gain $A_t \gamma_t^i / B_t(P_t)$ from (16). □

Proof of Proposition 3. The Bayesian agent has an objective perceptual set $\mathbb{P}$. The observable process P_t follows

$$\mathrm{d}P_t = V \,\mathrm{d}t + \sigma \,\mathrm{d}Z_t, \tag{18}$$

which has the solution $P_t = P_0 + Vt + \sigma \int_0^t \mathrm{d}Z_t$. At each t, the Bayesian agent compute the conditional mean and variance of V_t by $m_t = \tilde{\mathbb{E}}\left[V \mid \mathcal{F}_t^P\right]$ and $\gamma_t = \tilde{\mathbb{E}}\left[(V - m_t)^2 \mid \mathcal{F}_t^P\right]$, respectively. The innovation process becomes

$$\mathrm{d}Z_t^P = \frac{1}{\sigma}\left(\mathrm{d}P_t - m_t\,\mathrm{d}t\right). \tag{19}$$

It is a standard Wiener process with respect to $\mathcal{F}_t^P$ under $\mathbb{P}$. The Bayesian agent's belief updating process follows:

$$\mathrm{d}m_t = \frac{\gamma_t}{\sigma}\,\mathrm{d}Z_t^P, \tag{20}$$

$$\mathrm{d}\gamma_t = -\frac{(\gamma_t)^2}{\sigma^2}\,\mathrm{d}t\,. \tag{21}$$

By integrating (21), we obtain

$$\gamma_t = \frac{\gamma_0\sigma^2}{\gamma_0 t + \sigma^2}. \tag{22}$$

Since $\lim_{t\to\infty}\gamma_t = 0$, the unknown state V is eventually learned. To solve for the expected state of world, combine (19) and (20) to get

$$\begin{aligned}
\mathrm{d}m_t &= \frac{\gamma_t}{\sigma^2}\left(\mathrm{d}P_t - m_t\,\mathrm{d}t\right)\\
\implies \quad \frac{1}{\gamma_t}\,\mathrm{d}m_t &= -\frac{m_t}{\sigma^2}\,\mathrm{d}t + \frac{1}{\sigma^2}\,\mathrm{d}P_t\\
\implies \frac{1}{\gamma_t}\,\mathrm{d}m_t + m_t\,\mathrm{d}\frac{1}{\gamma_t} &= \frac{1}{\sigma^2}\,\mathrm{d}P_t\\
\implies \quad \mathrm{d}\left(\frac{m_t}{\gamma_t}\right) &= \frac{1}{\sigma^2}\,\mathrm{d}P_t\\
\implies \quad m_t &= \frac{\gamma_t m_0}{\gamma_0} + \frac{\gamma_t}{\sigma^2}\left(Vt + \sigma\int_0^t \mathrm{d}Z_s\right).
\end{aligned}$$

Plug in the solution of γ_t to finally arrive at

$$m_t = \frac{m_0\sigma^2}{\gamma_0 t + \sigma^2} + V\frac{\gamma_0 t}{\gamma_0 t + \sigma^2} + \frac{\gamma_0\sigma}{\gamma_0 t + \sigma^2}\int_0^t \mathrm{d}Z_s\,. \tag{23}$$

Then it is obvious to see that $\lim_{t\to\infty} m_t = V$. So the Bayesian agent eventually learns the true state of the world. □

Behavioral agent i views the world through the lens of her subjective perceptual set $\mathbb{P}^i$. P_t has the following dynamics under measure $\mathbb{P}^i$:

$$\mathrm{d}P_t = (V - \eta^i)\,\mathrm{d}t + \sigma\,\mathrm{d}Z_t^i, \tag{24}$$

which has the solution $P_t = P_0 + (V - \eta^i)t + \sigma\int_0^t \mathrm{d}Z_s^i$. At time t, from agent i's perspective, V has conditional mean

$$m_t^i = \tilde{\mathbb{E}}^i\left[V \mid \mathcal{F}_t^P\right] \tag{25}$$

and variance

$$\gamma_t^i = \tilde{\mathbb{E}}^i\left[\left(V - m_t^i\right)^2 \mid \mathcal{F}_t^P\right]. \tag{26}$$

The innovation process

$$\mathrm{d}Z_t^i = \frac{1}{\sigma}\left(\mathrm{d}P_t - (m_t^i - \eta^i)\,\mathrm{d}t\right). \tag{27}$$

It is a standard Wiener process with respect to $\mathcal{F}_t^P$ under $\mathbb{P}^i$. Behavioral agent i updates belief as follows:

$$\mathrm{d}m_t^i = \frac{\gamma_t^i}{\sigma}\,\mathrm{d}Z_t^i, \tag{28}$$

$$\mathrm{d}\gamma_t^i = -\frac{\left(\gamma_t^i\right)^2}{\sigma^2}\,\mathrm{d}t\,. \tag{29}$$

By integrating (29), we obtain

$$\gamma_t^i = \frac{\gamma_0\sigma^2}{\gamma_0 t + \sigma^2}, \tag{30}$$

which is independent from i and the same as that of the Bayesian agent. At a given time t, all behavioral agents and the Bayesian agent have the same posterior precision about V. To solve for the expected state of world, we combine (27), (28) and (29) to get

$$\mathrm{d}m_t^i = \frac{\gamma_t^i}{\sigma^2}\left(\mathrm{d}P_t - (m_t^i - \eta^i)\,\mathrm{d}t\right) \implies m_t^i = \frac{m_0\sigma^2}{\gamma_0 t + \sigma^2} + V\frac{\gamma_0 t}{\gamma_0 t + \sigma^2} + \frac{\gamma_0\sigma}{\gamma_0 t + \sigma^2}\int_0^t \mathrm{d}Z_s^i\,. \quad (31)$$

Equivalently, in observable-signal form,

$$m_t^i = \frac{m_0\sigma^2 + \gamma_0(P_t - P_0 + \eta^i t)}{\gamma_0 t + \sigma^2}.$$

Under the physical measure, this expression converges almost surely to $V + \eta^i$, consistently with Proposition 3.